

applications required by most of the users to access compatible desktop environments. Some popular package management distributions used in this segment are- OpenSuse, Slackware, Linux mint, Kubuntu, and Mageia.

- **Gnome**- this package management system is developed by the community, and especially for the community. It can easily be expanded as per the requirement with the help of its plugins. Gnome also doesn't require various resources, and it is a great choice for old hardware. The prominent distribution systems used in Gnome are- Debian, OpenSuse, Fedora and CentOS.

Installing, updating, removing repositories, dependencies, and packages

- **Installing**- even if the software is not packaged for any operating system officially, the developer or vendor will offer the package files for its download. In this case, you can easily use the retrieved files with your web browser or Via 'curl' on the command line. Once the package is established on the targeting system, then it can be installed with a single command.

On Debian systems, 'dpkg' can handle individual package files. If the package has unmet dependencies, then 'gdebi' can also be used to retrieve the file from official repositories. On Fedora and CentOS systems, 'yum' and 'dnf' can be used to install the individual files, and it can also handle required dependencies. Different systems have their different command lines which are mentioned below-

- Debian / Ubuntu- `sudo dpkg -i package.deb`
 - o `sudo apt-get install -y gdebi && sudo gdebi package.deb` (installation and using 'gdebi' to install package.deb while retrieving any other missing dependencies)
- Centos- `sudo yum install package.rpm`
- Fedora- `sudo dnf install package.rpm`
- FreeBSD packages- `sudo pkg add package.txz`
 - o `Sudo pkg add -f package.txz` (install package even if it is installed)

Updating- without any package system, it is tough to make sure that the entire installed software systems on the machine remains updated. It also is essential to track the upstream changes, and other security notifications for multiple packages. The package manager can solve small troubles or other problems that you may encounter while updating software. Upgrading the package can introduce various changes and require manual configuration. You have to read 'usr/ports/UPDATING' before upgrading with 'portmaster'. Some of the systems and their commands are mentioned below-